



Third Semester B.Tech (AI&ML/CCE) (CBCS) Degree Examinations

AM2101-1 – MODERN COMPUTER ARCHITECTURE
(For AI&ML, CCE)

Max. Marks:100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. The addressing mode/s, which uses the PC instead of a general purpose register is _____
A) Indexed with offset
B) Relative
C) Direct
D) Both Indexed with offset and direct
2. The condition flag Z is set to 1 to indicate _____
A) The operation has resulted in an error
B) The operation requires an interrupt call
C) The result is zero
D) There is no empty register available
3. The addressing mode, where you directly specify the operand value is _____
A) Immediate
B) Direct
C) Definite
D) Relative
4. The product of -13 and 11 is
A) 1100110011
B) 1101110001
C) 1010101010
D) 1111111000
5. A _____ gate is used to detect the occurrence of an overflow.
A) NAND
B) XOR
C) XNOR
D) AND
6. The delay reduced to in the carry look ahead adder is _____
A) 5
B) 8
C) 10
D) 2n
7. Carry look-ahead logic uses the concepts of _____
A) Inverting the inputs
B) Complementing the outputs
C) Generating and propagating carries
D) Ripple factor
8. The method used to reduce the maximum number of summands by half is _____
A) Fast multiplication
B) Bit-pair recoding
C) Quick multiplication
D) None of the mentioned
9. If the control signals are generated by combinational logic, then they are generated by a type of _____ controlled unit.
A) Micro programmed
B) Software
C) Logic
D) Hardwired
10. What is the primary advantage of pipelining in a CPU?
A) Reduced power consumption
B) Increased clock speed
C) Improved instruction throughput and reduced execution time
D) Greater cache memory capacity
11. In multiple Bus organization, the registers are collectively placed and referred as _____
A) Set registers
B) Register file
C) Register Block
D) Map registers
12. What does the hardwired control generator consist of?
A) Decoder/encoder
B) Condition codes
C) Control step counter
D) All of the mentioned

13. In microprogrammed control, what is a "microinstruction"?
 A) A small section of RAM B) A type of assembly language instruction
 C) A sequence of binary digits representing a control signal D) A subset of machine-level instructions
14. In multiple BUS organization _____ is used to select any of the BUSES for input into ALU.
 A) MUX B) DE-MUX
 C) Encoder D) None of the mentioned
15. What happens when a new instruction needs to be added or an existing one modified in a hardwired control unit?
 A) The CPU requires a complete redesign. B) The control unit's microcode is updated.
 C) A new hardware module is added. D) The control table is modified.
16. Coprocessors are often used to offload tasks from the main CPU, improving:
 A) CPU clock speed B) CPU memory capacity
 C) Overall system performance D) CPU power consumption
17. What is the difference between write through and write back?
 A) Write through refers to disk storage and write back refer to cache storage.
 C) Write through copies information to main memory immediately, but write back delays writing to memory until it's necessary.
 B) Write through refers to cache storage and write back refer to main memory storage.
 D) Write through saves writes until the CPU is through with the data, and write back keeps a copy so the CPU can get it back
18. CUDA stand for
 A) Compute Unified Device Architecture B) Compute Unit Device Architecture
 C) Compute Unified Disk Architecture D) Compute Uniq Device Architecture
19. What is CUDA?
 A) A programming language B) A parallel computing platform and API
 C) A type of graphics card D) A storage device
20. Which of the following is an advantage of using DMA?
 A) Slower data transfer rates compared to CPU-controlled transfers B) Increased CPU utilization during data transfers
 C) Reduced CPU overhead during data transfers D) Higher power consumption

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and briefly define the main structural components of a processor.	4	L*2	1	1
	b) What general categories of functions are specified by computer instructions? Illustrate with example.	4	L2	1	1
	c) List the types of transfers a computer's interconnection structure (e.g., bus) must support.	4	L2	1	1
	d) With neat sketches, explain various methods for handling multiple interrupts requests raised by multiple devices.	4	L2	1	1
2.	a) With state diagram briefly define the possible states that define an instruction execution.	5	L2	1	1
	b) Explain the following: i) Byte addressability ii) Big-endian assignment iii) Little-endian assignment	3	L2	1	1
	c) What is the effective address of the source operand in each of the following instructions, when the register R1 and R2 of computer contain the decimal value 1200 and 4600? i) Load 20(R1),R5 ii) Store R5, 30(R1,R2) iii) Subtract (R1)+,R5	3	L2	1	1
	d) Illustrate carry-save addition of summands by considering multiplication between 45 and 63 as an example.	5	L3	1	1

3. a)	Perform the division of $14 \div 3$ using non-restoring division.	5	L3	2	1
b)	Given the multiplicand $A = +23$ and multiplier $B = -10$. Perform the multiplication of A and B using Booth's algorithm.	5	L3	2	1
c)	Describe the principle of carry look-ahead addition for 4-bit adder circuit built using B-cells and calculate the number of gate delays for S3 and C4.	6	L3	2	1

Unit – II

4. a)	Draw and explain multiple bus organization of CPU. Explain its advantages.	6	L2	3	1
b)	Write and explain the control sequence for execution of the instruction Add (R3), R1.	6	L3	3	1
c)	Draw and explain organization of the control unit to allow conditional branching in the micro program.	4	L2	3	1
5. a)	What is cache memory? Explain different mapping functions with diagram.	10	L3	3	1
b)	Compare and contrast multiprocessors and multicomputers.	6	L3	3	1
6. a)	Explain cache coherence/cache consistency problem. How it can be overcome with write through cache coherence protocol and MESI Cache Coherence Protocol. Illustrate with example.	10	L2	4	1
b)	Discuss the super scalar approach to processor design with neat diagram.	6	L2	4	1

Unit – III

7. a)	How does direct memory access (DMA) work?	6	L2	5	4
b)	What is CUDA, and how does it enable general-purpose computing on GPUs?	5	L2	5	4
c)	Explain the concept of data parallelism with an example. Compare it with task parallelism.	5	L2	5	4
8. a)	What are the fundamental architectural differences between a CPU and a GPU, and how do these differences influence their respective functions.	7	L2	5	4
b)	Briefly define the following terms: i) True data dependency ii) Procedural dependency iii) Resource conflicts	4	L2	5	4
c)	What unique architectural elements define a GPU, and how do they enable the GPU's specialization in parallel processing.	5	L2	5	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
